HALCYON OPEN APPLICATION · 2026

RimaAI

An AI farming extension officer in every farmer's pocket.

Offline crop & livestock diagnosis, planting forecasts and outbreak alerts for Zimbabwe's smallholders, on smartphones *and* feature phones.

"Rima" (chiShona): to farm, to plough

rimaai

THE PROBLEM

Farmers lose harvests and herds to problems that are preventable. The missing piece is timely information.

100,000s

cattle lost

January disease (theileriosis) killed hundreds of thousands of cattle in 2017–2021. Dipping prevents it and medicine treats it, but farmers missed the schedules.

Too late

crop disease spotted

Maize lethal necrosis, tomato blight and fall armyworm are usually identified after the damage is done.

1 : 100s

officers to farmers

AGRITEX extension officers are stretched thin, so advice is rare and slow.

Offline

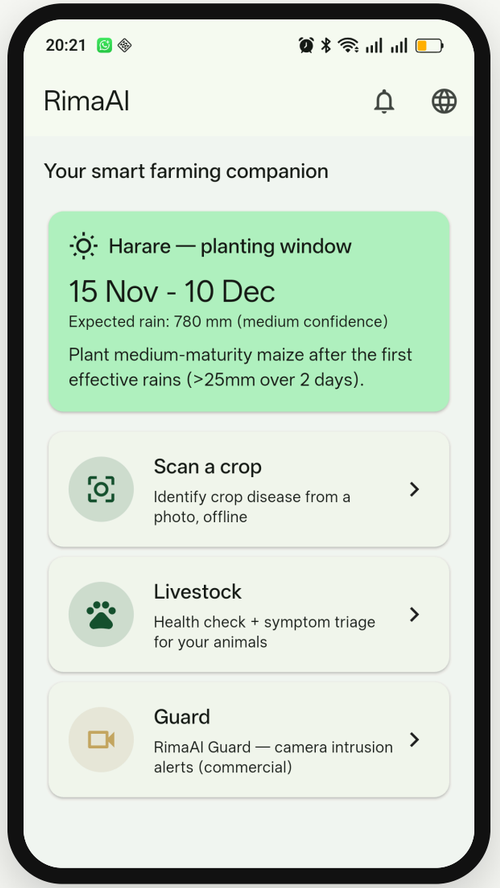
most of the time

Rural signal is patchy, data is expensive and most advice is English-only.

THE SOLUTION

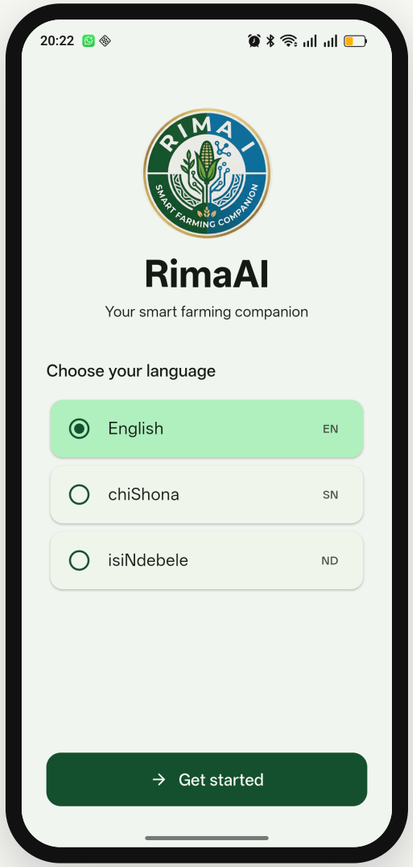
One multi-channel assistant that works where farmers are

- **Crop Scanner:** photo → disease → treatment advice, on-device and offline
- **Livestock Advisor:** photo health check and symptom triage chat that flags emergencies such as January disease
- **Planting forecasts:** district-level planting windows and rainfall guidance
- **Outbreak heat map:** community reports become district risk scores, which trigger SMS/WhatsApp alerts automatically
- **USSD *123#, SMS and WhatsApp** for feature phones, with no data needed
- **English · chiShona · isiNdebele** throughout

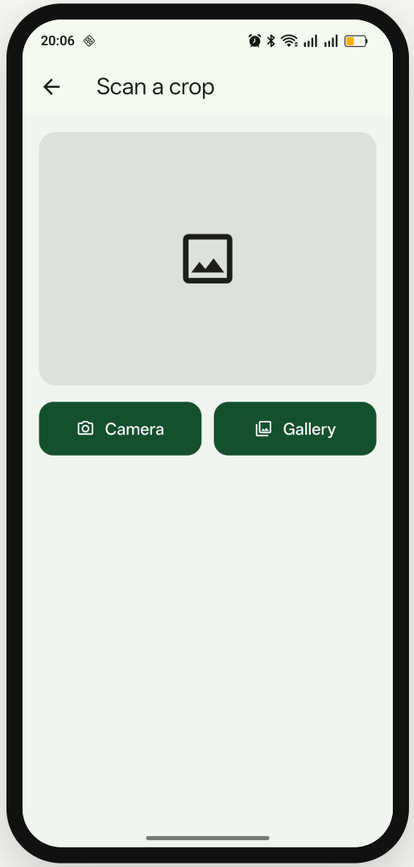


PRODUCT · MOBILE APP

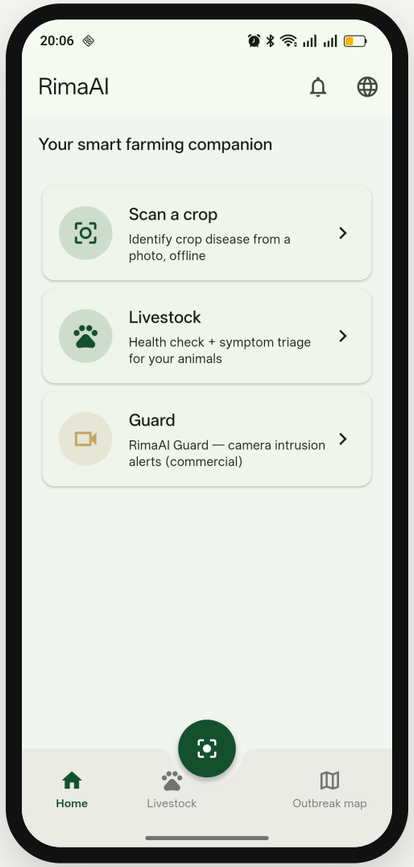
Built for low-cost Android phones and works offline



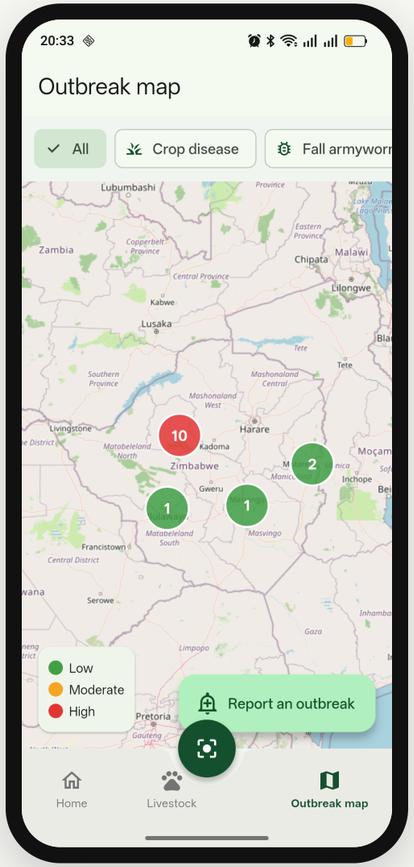
Pick a language



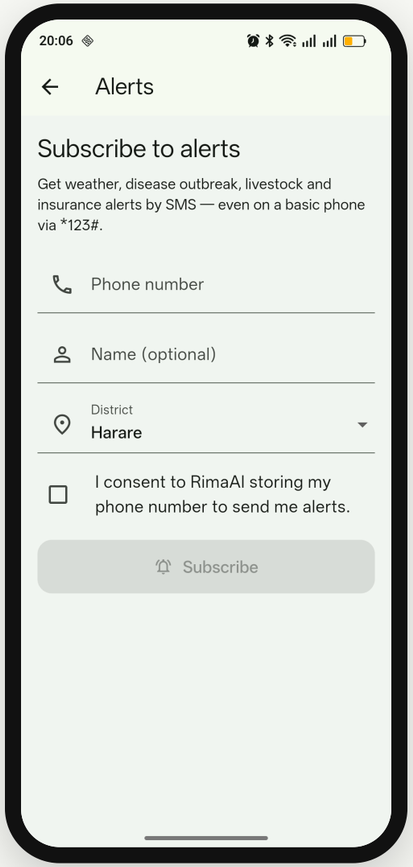
Scan a crop



Livestock triage



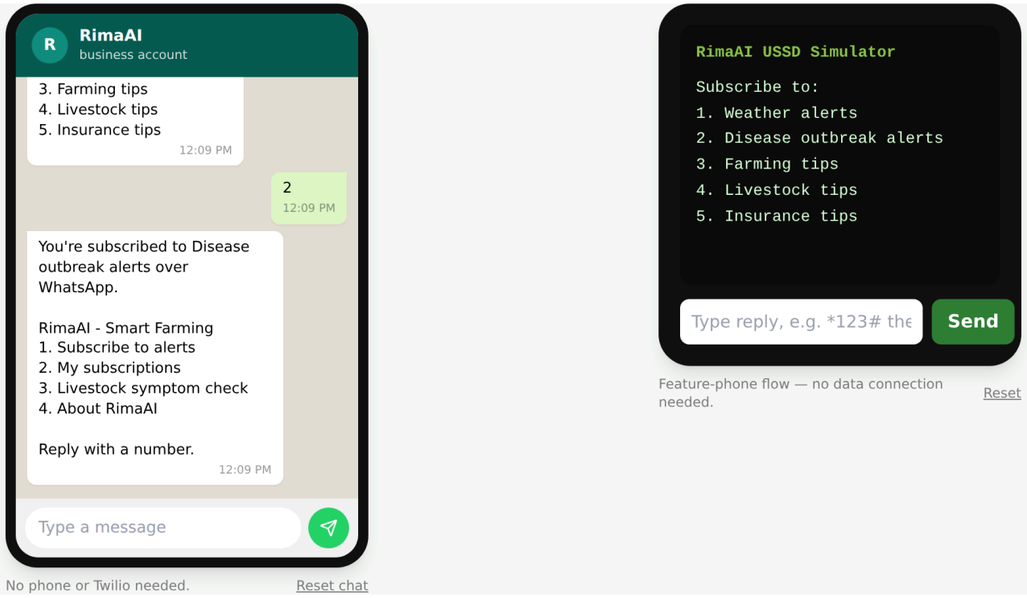
Outbreak heat map



Consent-based alerts

PRODUCT · EVERY PHONE

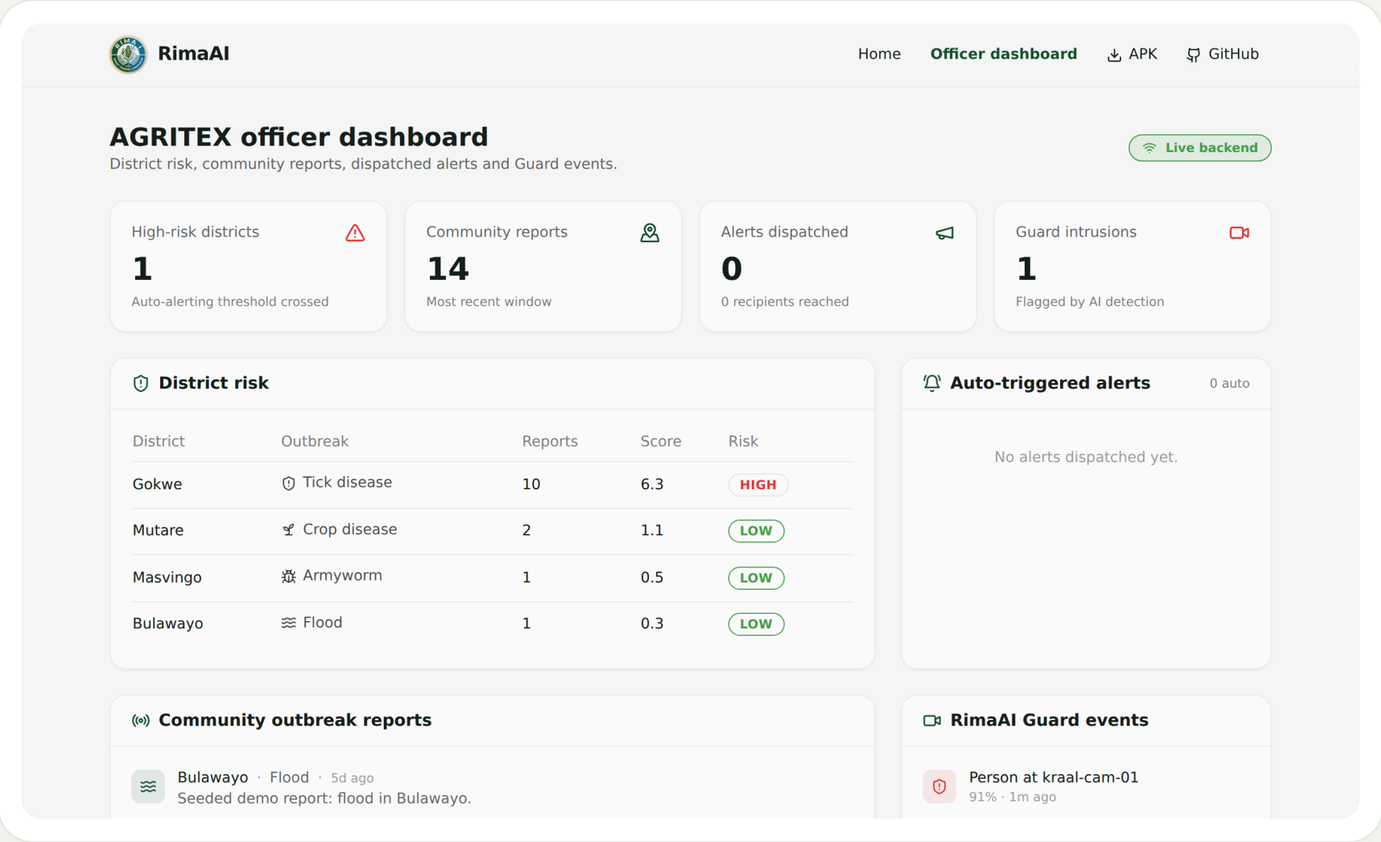
No smartphone or data bundle needed



- **USSD *123#:** subscribe to weather, outbreak, farming, livestock and insurance alerts from any handset
- **WhatsApp bot:** menu-driven subscriptions and livestock symptom checks
- **SMS:** dipping and vaccination reminders and outbreak warnings
- Twilio integration is code-complete

PRODUCT · WEB

AGRITEX officer dashboard: see outbreaks forming



- District risk is scored from community reports by **count × recency × trust**
- Once a district crosses the threshold, subscribers there get alerts automatically
- AI handles first-line triage, so officers only see flagged cases
- RimaAI Guard events (kraal-camera intrusions) appear on the same view

TECHNOLOGY

AI where it is needed, and plain rules elsewhere

Computer vision

MobileNetV3-Small, quantized to int8 TFLite, runs on the phone. Targets: under 100 ms and under 256 MB. Server fallback is available.

Language triage

Understands mixed Shona/English symptom text through an LLM, with a rule-based fallback. Keyword matching alone fails on paraphrased descriptions.

Class-aware detection

Guard tells people and vehicles apart from cattle moving at night, which simple motion detection cannot do.

Deliberately no AI

Subscriptions, alert scheduling and outbreak risk aggregation are transparent weighted rules that can be audited.

Stack

Flutter · FastAPI · Next.js · Supabase (Postgres + RLS) · Twilio · CI on every push · Dockerised for ZCHPC deployment

CUSTOMERS

Who we serve, and who pays

Users: smallholder farmers

Mixed crop and cattle farmers in rural Zimbabwe (e.g. Gokwe, Midlands, Masvingo) with 1–5 ha and 5–30 cattle. They speak Shona or Ndebele first, have little extension access, and use feature phones or low-cost Android phones.

Free core advice

Payers: institutions (B2B2C)

NGOs, agribusinesses, input suppliers, insurers and cooperatives fund access for their farmer networks. In return they get outreach channels, alerts and aggregated, anonymized field data.

Institutional subscriptions

Upsell: commercial farmers

Livestock owners who lose cattle to stock theft pay for RimaAI Guard, which sends camera intrusion alerts from the kraal.

Guard subscription

IMPACT

Measured against the UN SDGs

SDG 1 No Poverty **SDG 2 Zero Hunger** **SDG 11 Sustainable Communities** **SDG 13 Climate Action** **SDG 15 Life on Land**

Farmers reached (app, USSD, SMS, WhatsApp)	**Primary reach metric**
Crop loss among active users	**Target: 15–20% reduction**
Livestock disease caught early	**Triage escalations and vet referrals**
Income protected per farmer	**Losses avoided (crops and cattle)**
Farmers following planting-window advice	**Climate adaptation (SDG 13)**
Untargeted pesticide use avoided	**Land health (SDG 15)**

WHERE WE ARE

Working MVP across all channels; field pilot next

Built

- Flutter app (release APK)
- FastAPI backend with CI
- Officer dashboard (web)
- USSD and WhatsApp flows
- TFLite training pipeline

Next 6 months

- Collect local crop and livestock images with AGRITEX
- Train production models on Zimbabwean data
- Partner with MSD for weather data
- Pilot in a high-risk district

12–24 months

- Live telco USSD and SMS shortcode
- ML model to predict outbreak spread
- Payments for Guard and institutional tiers
- Expand to more districts

RESPONSIBLE BY DESIGN

Trust is the product

Privacy

Phone numbers are stored only with consent, with minimal personal data and row-level security. Complies with Zimbabwe's Data Protection Act [Chapter 11:12].

Human oversight

All advice is guidance only. Emergencies are escalated to a vet or AGRITEX officer.

Inclusive

Supports three languages, works offline and reaches feature phones, so no farmer is left out by device or data costs.

WHAT WE'RE ASKING HALCYON FOR

Help us take RimaAI from MVP to farmers in the field.

- Mentorship on B2B2C go-to-market with NGOs, agribusinesses and insurers
- Introductions to AGRITEX, telcos and impact investors
- Pilot funding for local data collection, model training and a district rollout

RimaAI: timely advice that saves crops and herds.

Thank you